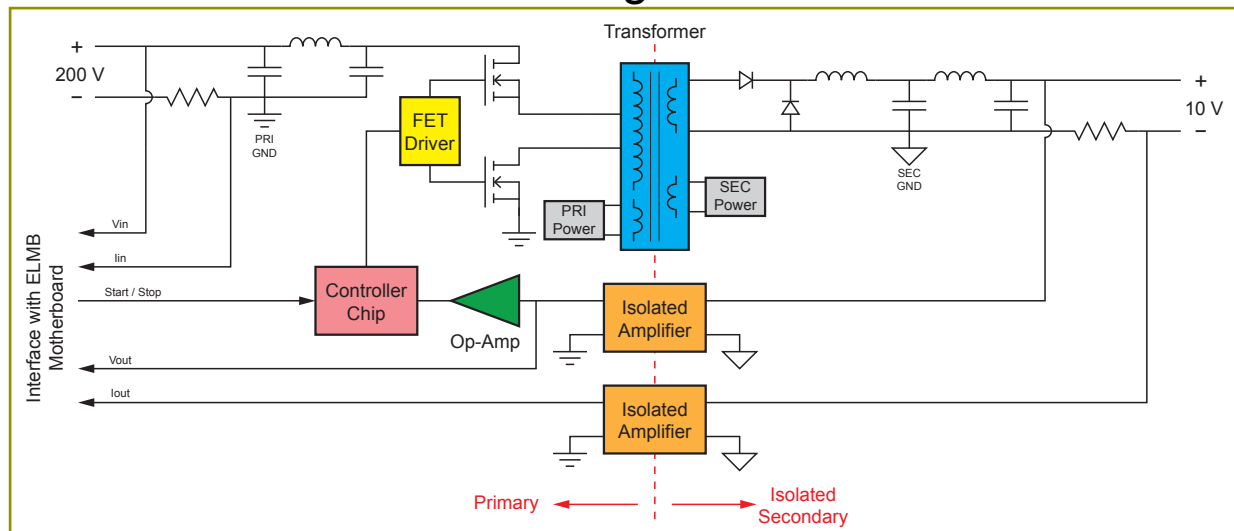


Tile Calorimeter Power Supplies: LVPS Bricks

- The LVPS bricks are switching DC-DC converters with a topology of two-switch forward converter and a switching frequency of 300 kHz
- In the legacy system, 8 different bricks inside the box provide power to each module of the Tile Calorimeter at 8 different voltage levels, but in the phase-2 upgrade system (for HL-LHC), all 8 bricks in a module will convert the 200 V input to a nominal 10 V output at a nominal power of 23 W
 - Point of load regulators on front-end electronics provide different voltages
- The phase-2 system improved efficiency and redundancy of the system, as a module will still be able to read out all the cells (half the PMTs) even with 4 out of 8 bricks running



LVPS brick functional block diagram



LVPS brick top view